

Patchy Environments I

Today's Agenda:

- Quiz
- Group Assignment
 - Project 4
- Course Eval Reminder
- Patchy Environments

Group Assignment

- Focus: variable/changing ecosystems in Florida
- Variation due to:
 - Ecological succession
 - Temporally fluctuating environments
 - Global changes
- Some obvious examples might be:
 - Fire-dependent ecosystems
 - Tidal systems
 - Hurricanes as successional drivers
 - Ephemeral water bodies

Group Assignment

- Pick your own group (Groups can be of one person)
- Everyone in a group gets the same grade
- Do/make something that explains your your chosen system to lay people
- Can be anything that can convey a message
 - Possibly bonus points for creativity

Group Assignment

- Products will be presented in class on April 29th
- Show off what you created to the class (and me)
- Your group should briefly explain:
 - what you did/created
 - rationale behind the approach you took
 - how it conveys the underlying concepts

Questions so far?

Project 4

Due May 8th before midnight

- University policies force me to be a bit stricter about the deadlines on this one
- Feel free to submit early to avoid the end-of-semester crunch!

Project 4

- Pick a research question about community ecology
- Propose alternative research approaches
- Three parts:
 - 1) Intro (1-2 pages, 4 max)
 - 2) Methods (**funded** version, 1-2 pages)
 - 3) Methods (**free** version, 1-2 pages)

Project 4

- Ecology is cool because it can often be done for free
- BUT you can usually do even better work with some funding
- Goal here is to think about different approaches to a question

Project 4

- No set number of references to read
- Re-use as much old text as you like
- Remember to cite where you need to
- Worth 15 points (other projects were 10)

Questions on Project 4?

Course Evaluations

- Going out soon apparently
- Feedback is helpful (particularly for a new course like this)

Patchy Environments

Patches: suitable habitat, surrounded by unsuitable habitat.

Metapopulation: collection of connected populations

- Multiple patches connected by dispersal

Levins Metapopulation Model

Assumptions:

- Environment composed of identical patches
- Patches connected by dispersal
- Patches are either at carrying capacity or empty
- Patches have a constant extinction rate
- Colonization rate depends on number of patches occupied

Levins Metapopulation Model

Model:

c = per-patch colonization rate

P = Proportion occupied patches

m = per-patch extinction rate

$$dP/dt = cP(1 - P) - mP$$

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$$mP = cP(1 - P)$$

$$m/c = (1 - P)$$

$$P = 1 - m/c$$

Levins Metapopulation Model

Model:

c = per-patch colonization rate

P = Proportion occupied patches

m = per-patch extinction rate

$$dP/dt = cP(1 - P) - mP$$

$$0 = cP(1 - P) - mP$$

$$mP = cP(1 - P)$$

$$m/c = (1 - P)$$

Patches occupied at equilibrium

$$P = 1 - m/c$$

Levins Metapopulation Model

Model:

c = per-patch colonization rate

P = Proportion occupied patches

m = per-patch extinction rate

At equilibrium:

$$P = 1 - m/c$$

M decreases occupancy

C increases occupancy

Levins Metapopulation Model and habitat loss

Model:

c = per-patch colonization rate

P = Proportion occupied patches

m = per-patch extinction rate

D = Proportion of patches destroyed

$$dP/dt = cP(1 - D - P) - mP$$

Levins Metapopulation Model and habitat loss

Model:

c = per-patch colonization rate

P = Proportion occupied patches

m = per-patch extinction rate

D = Proportion of patches destroyed

$$dP/dt = cP(1 - D - P) - mP$$

at $D = 1 - m/c$

Extinction occurs

Levins Metapopulation Model and habitat loss

Model:

c = per-patch colonization rate

P = Proportion occupied patches

m = per-patch extinction rate

D = Proportion of patches destroyed

$$dP/dt = cP(1 - D - P) - mP$$

$$\text{at } D = 1 - m/c$$

Extinction occurs

Note that we can make extinction harder by increasing colonization or decreasing patch-level extinction

Levins Metapopulation Model and habitat loss

Other implications:

- Empty patches are always expected
- Even destroying an unoccupied patch can cause extinction
- Extinction may not occur immediately
 - Occupied patches may persist, even if metapop can't persist
 - "Extinction debt"
 - BIG concern in conservation biology

Levins Metapopulation Model and dispersal

Dispersal corridors:

- Increase colonization (reducing extinction)
- Meta-analyses suggest 50% increase in colonization

Conservation Biology 

Review

A Meta-Analytic Review of Corridor Effectiveness

LYNNE GILBERT-NORTON,* RYAN WILSON,* JOHN R. STEVENS,† AND KAREN H. BEARD‡§

Improving dispersal

How else might we improve dispersal?

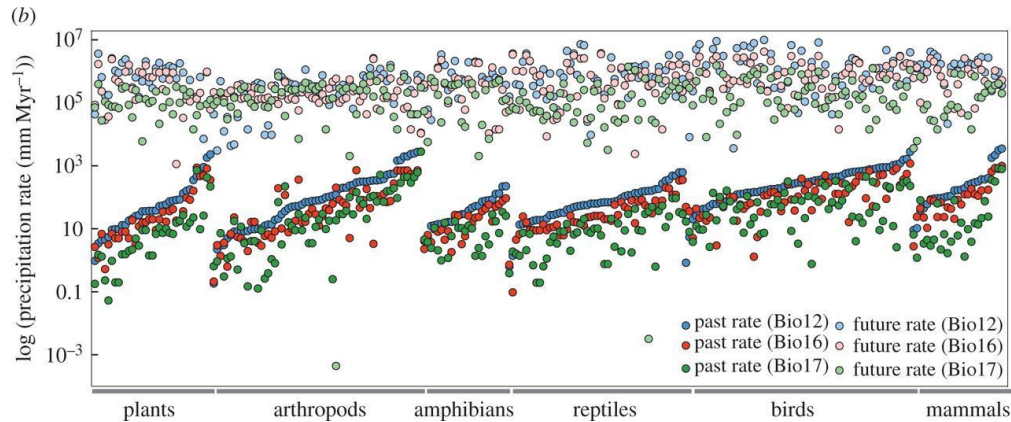
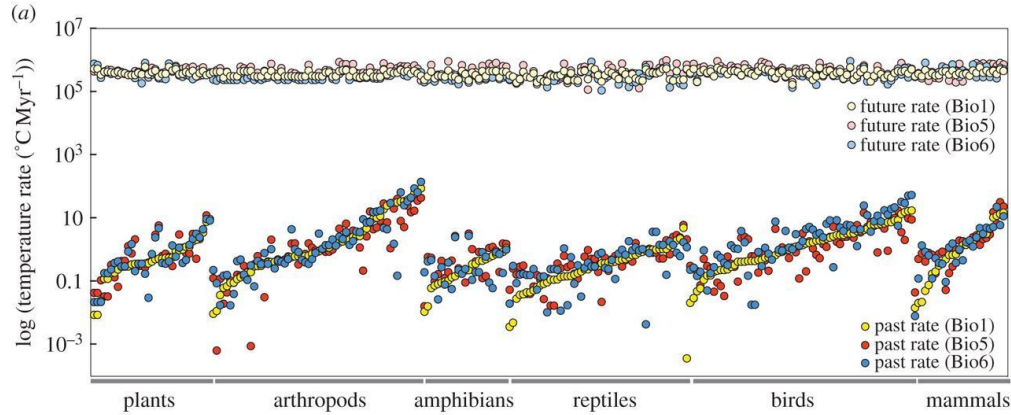
Assisted Migration

AKA assisted dispersal, assisted colonization

Rationale:

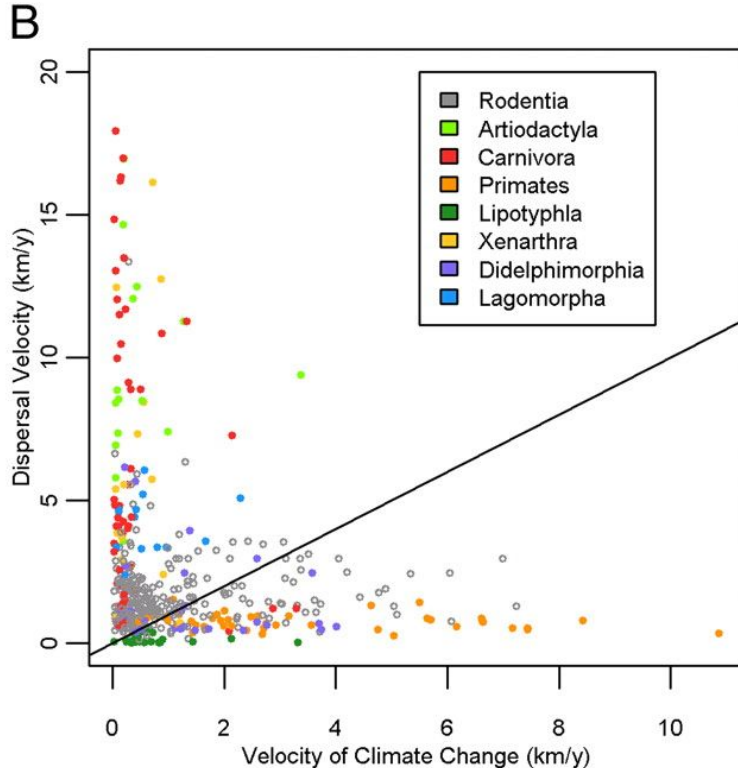
- Environment changing faster than species can disperse naturally
- Even if they could disperse fast enough, barriers may hinder this
- We can help disperse species where needed

Assisted Migration



Niche evolution is much slower than projected climate change

Assisted Migration



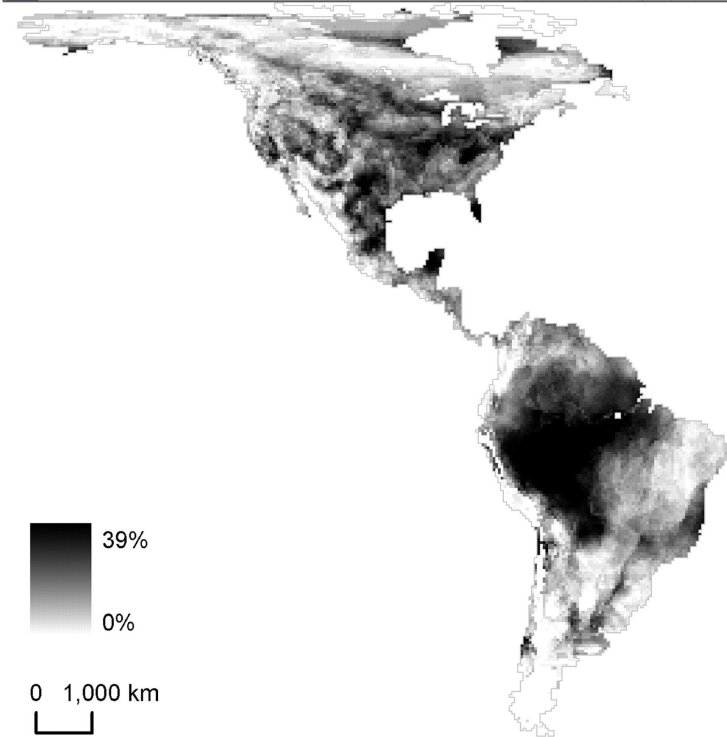
Dispersal will limit ability of mammals to track climate change in the Western Hemisphere

Carrie A. Schloss¹, Tristan A. Nuñez, and Joshua J. Lawler

School of Environmental and Forest Sciences, University of Washington, Seattle, WA 98195

Edited by Monica G. Turner, University of Wisconsin, Madison, WI, and approved April 13, 2012 (received for review October 12, 2011)

As they have in response to past climatic changes, many species will shift their distributions in response to modern climate change. However, due to the unprecedented rapidity of projected climatic changes, many species may not be able to track climate change. The ability of species to track climate change can be measured as the distance from that location within the species' current range to the closest location projected to have a suitable climate in the future divided by the time period



Species unable to track climate

Assisted Migration

Will plant movements keep up with climate change?

Richard T. Corlett¹ and David A. Westcott²

Mean rate of temperature change: 0.42 km/yr

Mean of rate of precipitation change: 0.22 km/yr

Plant dispersal (high estimates) 1.7 m/yr - 1.5 km.yr

- BUT actually likely lower because long dispersal is correlated with large size (long generations)

Most plants won't be able to keep up

Assisted Migration

- Sometimes opposed due to parallels with introduced species
- Runs counter to managing ecosystems by NOT changing them

Will plant movements keep up with climate change?

Richard T. Corlett¹ and David A. Westcott²

Will plant movements keep up with climate change?

Richard T. Corlett¹ and David A. Westcott²

Assisted Migration

- Sometimes opposed due to parallels with introduced species
- Runs counter to managing ecosystems by NOT changing them

BUT:

- Introductions aren't usually bad
- These aren't being introduced into entirely new communities
- Traits identifying a candidate for A.M. are the opposite of those of an invasive species
 - Slow spread, low population densities, etc.
- The real concerns are with logistics

Metapopulation models and epidemiology

- Metapopulation models similar to infection models
 - Hosts = patches
 - Infection = colonization
 - Recovery = patch extinction
 - Host death = patch destruction
 - Vaccination = patch destruction

Metapopulation models and epidemiology

- Diseases require a sufficient number of infections to persist
 - Comparable to metapopulation dynamics
 - Just need to vaccinate $1-m/c$ proportion of people!

Metapopulations and restoration

Eastern collared lizards (*Crotaphytus collaris collaris*)

- Live in dry, open glades (maintained by fire)
- Fire suppression eliminated glades
- In 1994 prescribed burns started



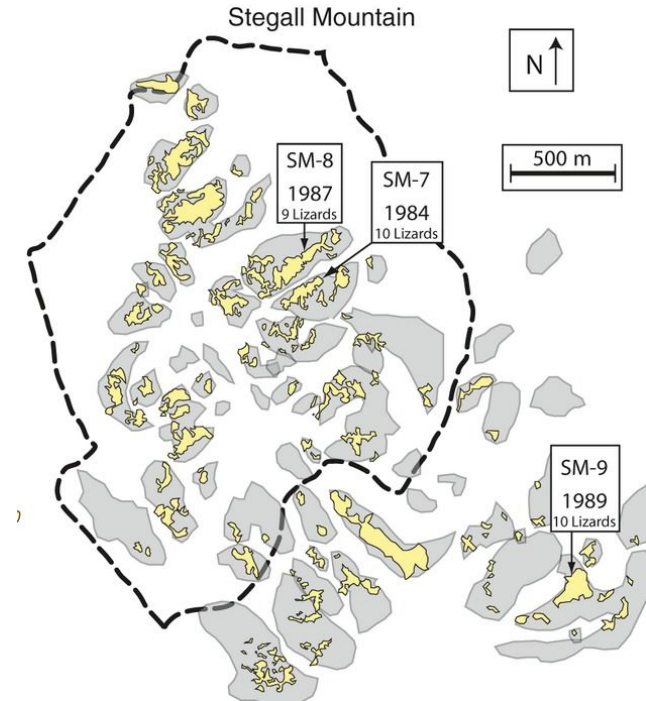
NEWS AND VIEWS

PERSPECTIVE

Extinction, reintroduction, and restoration of a lizard meta-population equilibrium in the missouri ozarks

JACK W. SITES JR

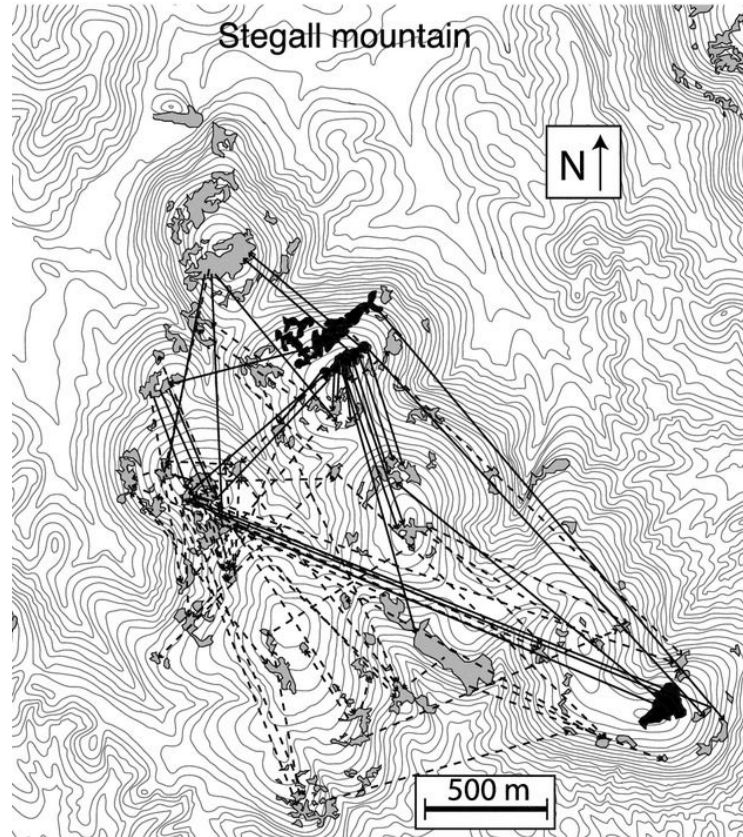
The Missouri forest in w ridge tops a ical of sout scorpions, lared lizard populations



Metapopulations and restoration

Genetic restoration in the eastern collared lizard under prescribed woodland burning

JENNIFER L. NEUWALD*‡ and ALAN R. TEMPLETON*†

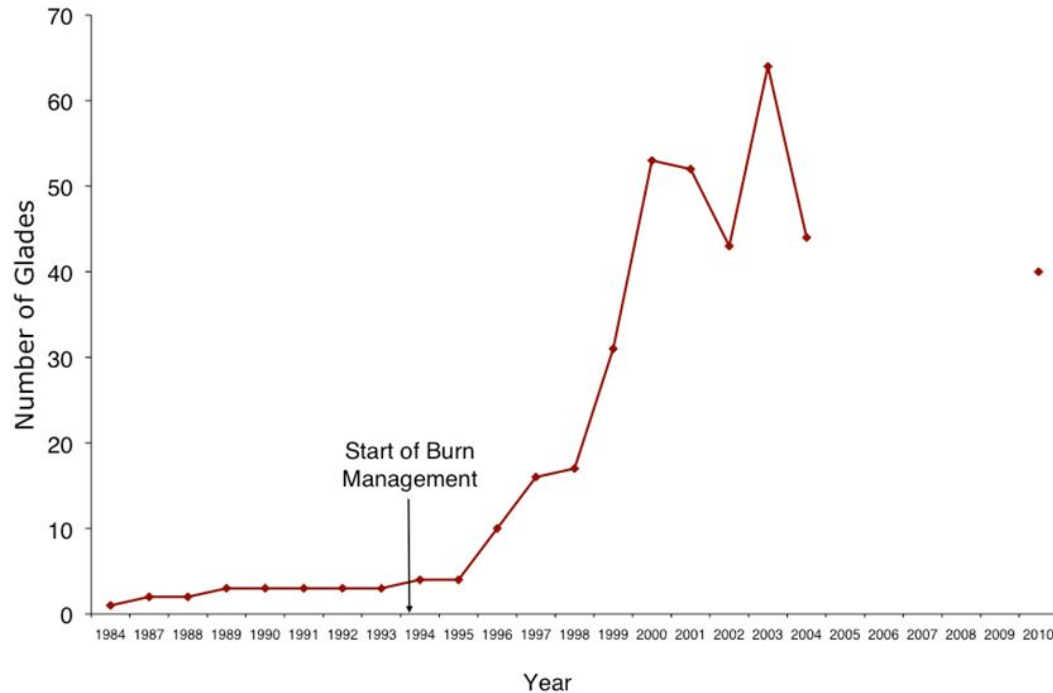


Solid lines = dispersal 1994–1998
Dashed lines = dispersal 1999–2000

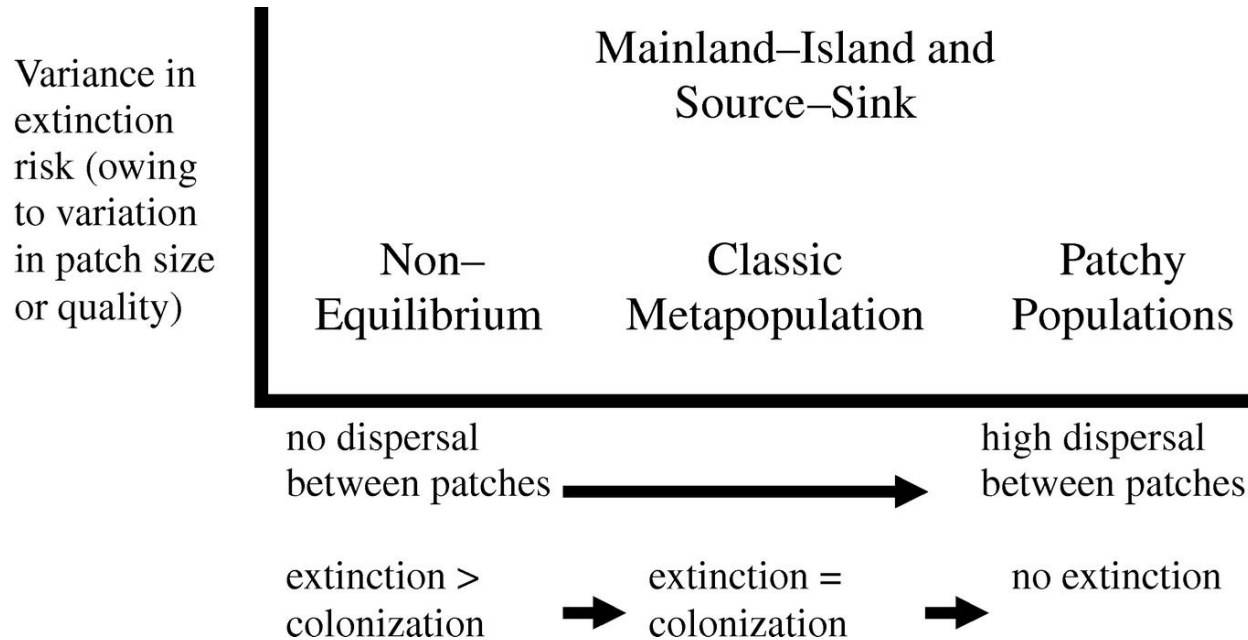
Metapopulations and restoration

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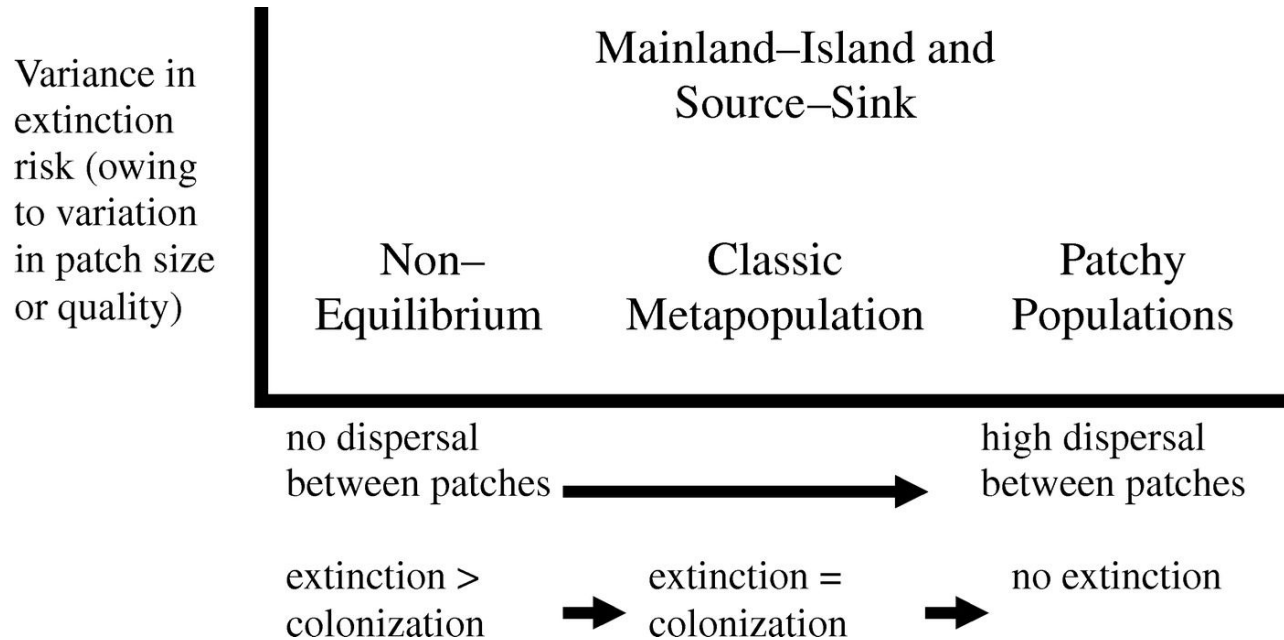
JENNIFER L. NEUWALD*‡ and ALAN R. TEMPLETON*†



Types of metapopulation structure



Types of metapopulation structure



**Important
conservation
implications!**

Next class: More patchy environments